

SIERRA VISTA, AZ, USA

WMO: 722730

Lat: 31.588N

Lon: 110.344W

Elev: 4719

StdP: 12.36

Time Zone: -7.00 (W07)

Period: 06-19

WBAN: 03124

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 27.4	(c) 30.5	(d) -3.0	(e) 5.6	(f) 56.5	(g) 0.5	(h) 6.7	(i) 58.7	(j) 30.3	(k) 50.7	(l) 25.9	(m) 55.4	(n) 5.0	(o) 260	(p) 0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 26.1	(c) 96.9	(d) 60.6	(e) 94.5	(f) 61.0	(g) 92.3	(h) 60.8	(i) 68.7	(j) 80.9	(k) 67.9	(l) 79.9	(m) 67.1	(n) 79.2	(o) 10.6	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	114.5	71.1	64.7	109.4	71.0	63.7	105.3	71.2	35.7	80.9	34.9	78.8	34.2	78.6	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.5	(b) 21.2	(c) 18.5	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 25.5	(b) 21.2	(c) 18.5	(d) WB	(e) 19.4	(f) 70.7	(g) 2.0	(h) 1.0	(i) 17.9	(j) 71.4	(k) 16.8	(l) 72.0	(m) 15.6	(n) 72.5	(o) 14.1	(p) 73.3	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	47.2	51.7	58.2	63.4	69.4	80.3	79.0	77.7	73.6	66.5	57.1	48.9
	DBStd	12.77	6.76	7.71	6.76	5.99	6.01	4.30	4.14	3.73	4.10	5.70	6.89	6.64
	HDD50	339	131	68	15	2	0	0	0	0	0	1	22	100
	HDD65	2067	551	374	222	96	30	0	0	0	1	51	243	499
	CDD50	5622	45	115	270	403	602	908	898	860	707	512	236	66
	CDD65	1874	0	1	12	47	167	458	433	395	258	96	7	0
	CDH74	20252	0	57	272	731	2078	5927	4315	3619	2170	947	132	4
	CDH80	8782	0	3	27	146	780	3230	2016	1570	774	225	11	0
(14) Wind	WSAvg	7.8	6.7	8.1	9.1	9.9	9.6	9.0	7.4	6.4	6.6	7.1	6.8	6.7
(15) Precipitation	PrecAvg	13.3	1.0	0.6	0.5	0.3	0.2	0.6	3.5	3.4	1.5	0.6	0.5	0.7
	PrecMax	19.8	6.0	1.7	2.5	1.5	1.5	3.7	5.6	5.9	5.6	6.8	3.1	3.0
	PrecMin	8.1	0.0	0.0	0.0	0.0	0.0	0.0	1.5	1.5	0.5	0.0	0.0	0.0
	PrecStd	3.6	1.3	0.6	0.6	0.4	0.3	0.8	1.3	1.1	1.0	1.3	0.6	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	79.9	83.1	86.9	92.3	101.3	98.3	96.2	93.0	88.3	81.5	73.5
		MCWB	47.7	49.9	51.1	53.1	55.2	61.1	61.9	63.9	61.3	58.0	52.0	47.9
	2%	DB	68.0	75.3	80.2	83.7	90.1	98.5	95.3	93.5	89.9	85.1	77.9	69.6
		MCWB	45.9	47.5	50.4	52.1	54.6	59.7	63.0	64.3	61.3	56.9	51.1	47.4
	5%	DB	64.2	71.3	77.2	81.4	87.5	96.0	92.9	91.1	87.4	82.4	74.6	65.8
		MCWB	44.6	46.4	49.3	50.4	53.9	59.1	63.5	64.4	61.2	56.5	50.4	46.0
	10%	DB	60.4	66.9	73.5	78.6	84.8	93.6	90.3	88.5	84.7	79.7	71.3	61.5
		MCWB	43.2	44.4	48.0	50.0	53.1	59.0	63.7	64.5	60.9	55.3	49.6	44.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	52.5	53.6	55.8	59.3	66.5	70.0	70.5	68.0	63.8	55.9	53.0
		MCDB	63.8	64.3	77.3	80.8	82.9	82.4	83.1	83.5	78.0	75.4	67.6	60.6
	2%	WB	49.0	50.4	51.5	53.7	57.1	64.8	68.7	69.1	66.4	61.7	53.5	50.2
		MCDB	60.2	65.1	74.5	78.1	80.7	83.3	80.8	81.3	77.3	76.3	70.3	62.7
	5%	WB	46.9	48.3	50.3	52.1	55.4	63.5	67.8	68.2	65.5	60.0	52.2	48.1
		MCDB	58.7	65.6	73.2	77.4	81.8	84.1	79.8	80.4	77.1	74.8	69.6	61.4
	10%	WB	45.0	46.3	48.9	50.6	53.9	62.2	67.0	67.4	64.4	58.4	50.6	46.1
		MCDB	57.4	64.9	71.1	75.7	81.9	84.6	79.1	79.6	75.9	73.3	68.1	59.1
(35) Mean Daily Temperature Range	5% DB	MDBR	23.2	25.0	26.3	26.6	26.1	26.1	21.1	20.9	21.5	24.4	24.5	22.9
		MCDBR	27.6	29.8	29.7	29.5	28.8	27.4	24.2	24.3	24.2	26.2	27.6	27.6
		MCWBR	13.3	13.1	13.0	12.7	11.4	10.4	6.6	6.9	7.7	10.3	12.0	13.3
	5% WB	MCDBR	22.0	25.0	27.4	27.1	26.7	24.5	19.8	19.6	19.1	22.7	24.3	22.9
		MCWBR	11.8	11.5	12.7	12.5	11.5	8.9	5.5	6.4	6.7	8.9	11.2	12.2
(40) Clear-Sky Solar Irradiance	taub		0.243	0.246	0.261	0.270	0.285	0.309	0.361	0.347	0.313	0.277	0.253	0.242
	taud		2.692	2.684	2.608	2.566	2.560	2.498	2.474	2.500	2.578	2.632	2.660	2.683
	Ebn at Noon		321	328	326	324	317	307	292	295	304	312	314	317
	Edh at Noon		23	25	29	32	32	34	35	34	30	26	23	22
(44) All-Sky Solar Radiation	RadAvg		1177	1475	1945	2376	2603	2536	2044	1982	1846	1618	1296	1081
	RadStd		80	119	108	83	101	148	141	77	91	101	64	49

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A

Nomenclature: See separate page